

POLICY IMPLICATIONS OF UNITED STATES (US) LIQUEFIED NATURAL GAS (LNG)

EXPORTS FOR SINGAPORE

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1. Introduction

The recent unconventional natural gas boom and the consequent gas glut in the United States (US) have the potential to fundamentally transform the dynamics of the global natural gas. Recent price differentials between Asian and North American gas are wide. Prices for contracted LNG imports into Japan have commonly exceeded US\$14/million British thermal units (MMBtu) while natural gas prices at Henry Hub have hovered around US\$4/MMBtu. Due to this price disequilibrium between oil-linked Asian contract supply and low-cost North American gas, there exists significant potential for growth in US LNG exports to Asia.

This paper will focus on policy implications of increased US LNG exports to Singapore. As an energy importing country, Singapore faces two major uncertainties, namely supply security and price fluctuations. Singapore relies on natural gas piped from Malaysia and Indonesia to generate 80 percent of its electricity and fuel its industry. However, both Malaysia and Indonesia are experiencing decreasing supply from ageing fields and increasing domestic natural gas demand. The growth in US LNG exports to Asia could provide alternative supply and diversification of energy portfolio for Singapore. This is supported by the LNG re-gasification and storage terminal which will be ready to operate by the second quarter of 2013.

The pricing formulas for most of the Asian LNG contracts tie natural gas prices to the price of crude oil which has been high in recent years. Thus, Japan, South Korea, Singapore and other Asian buyers pay the highest worldwide prices for natural gas. Additional exports of LNG from the United States are expected to reduce Asian gas prices. This, in turn, will have implications on the price that Singapore pays for its gas imports.

Nevertheless, regulatory hurdles may handicap the prospects of the United States becoming a significant exporter of LNG. There are major debates in the United States surrounding the impact of unconventional natural gas development on the environment as well as the impact of LNG export on domestic prices have been growing. The possibility of a jump in North American gas prices may also

undermine the economic feasibility of US LNG exports. Nevertheless, the jump in Henry Hub prices has to be substantial for this to happen.

2. Demand and Supply Factors

International gas market can be divided into three major regions: North America, the Pacific Basin which comprises Asian countries and the Atlantic Basin which comprises European countries. These three markets are supplied by liquefied natural gas (LNG) shipments and pipeline gas. Each market has different demand and supply characteristics. Nevertheless, both demand and supply factors in North America and the Pacific Basin have contributed to a scenario where significant growth in LNG exports from the former region to the latter region has become a real possibility.

2.1. Demand side

Natural gas demand is expected to remain strong in the Asia Pacific for the next 25 years, led by Japan, China and India. According to the International Energy Agency (IEA), Asia's demand for natural gas is projected to increase from 400 billions of cubic meters (bcm) in 2010 to 1060 bcm in 2035, an annual average compound growth rate of 4% (International Energy Agency, 2011c). Japan is the world's largest importer of LNG in Asia. The Fukushima nuclear power plant accident after the earthquake in March 2011 has led to deteriorating public confidence regarding Japan's reliance on atomic power and caused a shutdown of most of Japan's nuclear reactors. Japan will increase its natural gas demand to meet its electricity production in the event of a move away from nuclear power generation.

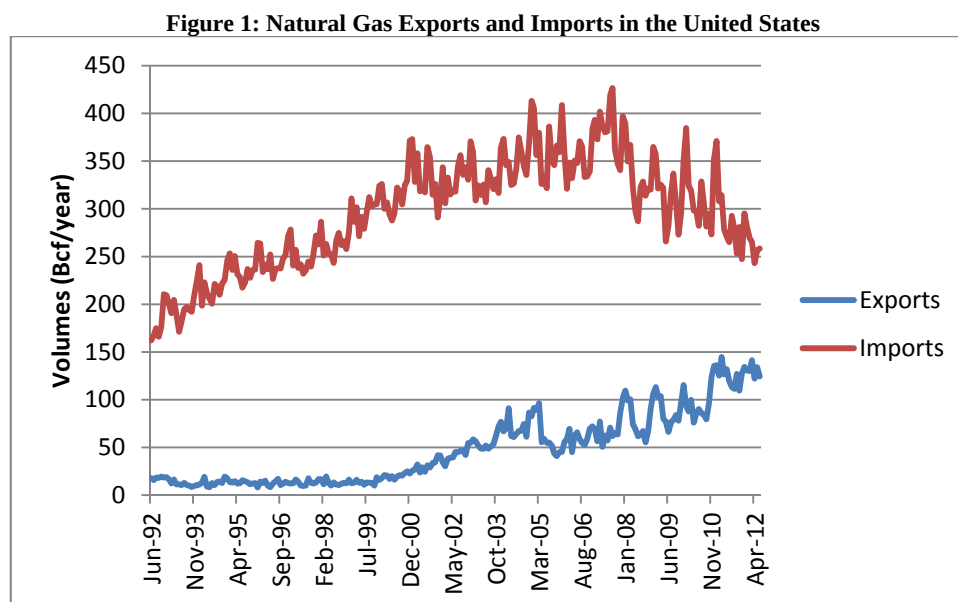
The economic rise of China and India will also result in increasing natural gas consumption in Asia. The IEA projects that gas demand in both countries will grow at a rate of between 6-7% per annum (International Energy Agency, 2011a). While coal and oil may still dominate energy supply and electricity generations in China and India, demand for natural gas is still projected to increase steadily. Both countries have been making significant investments in LNG import terminals and are looking to develop their pipeline infrastructure to improve their interconnection with their neighboring countries. China and India possess significant unconventional gas resources in form of shale gas and coal-bed methane (CBM) of their own. However, development of these resources requires technical capacity and physical infrastructure which they lack. Domestic unconventional gas production is likely to be

started in the medium to long term, and until then, their demand for natural gas imports is likely to keep growing.

2.2. Supply side

The growing demand in Asia Pacific consumption will be accompanied by a large increase in North American gas production. Less than a decade ago, the United States (US) faced a major shortfall in the supply of natural gas due to declining conventional production and rising demand. It has been exporting modest quantities of natural gas via pipeline to Mexico and Canada and a relatively small amount of its Alaskan liquefied natural gas (LNG) to Japan while importing natural gas via pipeline from Canada and majority of its LNG from Trinidad and Tobago, Qatar and Algeria.

The recent unconventional natural gas boom and the consequent gas glut in North America have the potential to transform the country into one of the world's largest producers and hence become a net exporter of natural gas. Figure 1 below illustrates how the shale gas revolution has led to falling imports and rising exports since 2005. Exports have been growing consistently in the 2000s, whereas imports rose in the first half of the 2000s before falling in the second half of the year.



Source: EIA natural gas databases, 2012 from <http://www.eia.doe.gov/naturalgas/data.cfm>.

US natural gas exports are driven by technological breakthroughs in unconventional gas production which have led to increased natural gas reserves and production. Further supply growth is projected in

the future on the basis of the potential for extraction of these vast reserves of unconventional gas. Unconventional gas from shale is produced by drilling horizontal wells (laterals) and conducting a hydraulic fracturing (or fracking) process in which water, sand and chemicals are forced into rock formations in high pressures to free trapped gas.

According to the International Energy Agency (IEA), North America accounted for around 11 % of the world's technically recoverable conventional gas resources at the end of 2011, but the figure rises to over 16 % once unconventional gas resources are included as well (International Energy Agency, 2012). Due to improved techniques in extracting natural gas from shale formations, several agencies like ICF International, Advanced Resources International, US Energy Information Administration (EIA) and Potential Gas Committee (PGC) have also confirmed the abundance of US natural gas in their resource assessments. Estimates of the size of recoverable unconventional gas resource have varied between under 700 trillion cubic feet (tcf) to over 1800 tcf (Ebinger et al., 2012). In 2010, the United States consumed around 24 tcf of gas, which means the estimates of unconventional gas alone will satisfy US domestic demand for around 30 to 80 years.

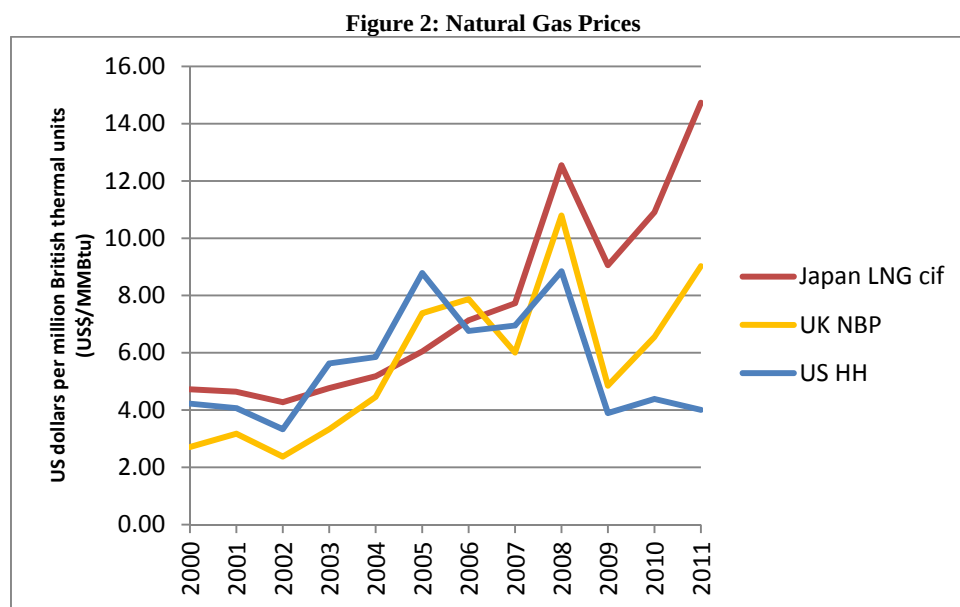
As a result, there has been increased interest in using natural gas in the US domestic sectors and in finding new sources of demand for the newfound volumes. The second interest focuses on the export of US natural gas in the form of liquefied natural gas (LNG). The liquefaction capacity in the United States could exceed 12.6 billion cubic feet per day (bcf/day) (Ebinger et al., 2012) if all of the proposed export projects go through. To put this number into context, the liquefaction capacity of Qatar (which currently has the largest capacity in the world) is 10.7 bcf/day (Energy Information Administration, 2011).

2.3. Price Differentials

As mentioned above, the global natural gas market is separated into three markets because of differences in market and pricing structure. Natural gas in the North America is traded in open market with a competitive and transparent price. The Atlantic Basin is a hybrid market which includes a competitive United Kingdom (UK) market and a rigid Continental European market. The US and UK

central trading hubs are called the Henry Hub (HH) and National Balancing Point (NBP) respectively. The Pacific Basin is dominated by oil-linked contracts. Japan and South Korea import LNG under pricing formulas based on the Japan Crude Cocktail (JCC) price, which is the average price of oil imports into Tokyo.

Recent price differentials in the major markets like Henry Hub and Japan are wide and they form the basis for the economic feasibility of US LNG exports to Asia. Figure 2 below shows that while natural gas prices at Henry Hub have hovered around the US\$4.00/million British thermal units (MMBtu) in the past several years, prices for contracted LNG imports into Japan (calculated as cif = cost + insurance + freight) have exceeded US\$14/MMBtu during the same period. This high oil-indexed priced natural gas in the Asian market is caused by the rising natural gas demand in Asia Pacific. Thus, at current price differentials, it makes economic sense for gas producers in the United States to supply LNG to the Asia-Pacific region. The break-even export price is approximately US\$9.90/MMBtu (Doshi et al., 2012), which is considerably less than the current Japanese contract price of US\$15.00/MMBtu.



Source: BP Statistical Review of World Energy, 2012 from <http://www.bp.com/sectiongenericarticle800.do?categoryId=9037181&contentId=7068643>.

3. Policy implications for Singapore

3.1. Supply Security via Diversification

Natural gas accounts for 80% of Singapore's power generation mix and the country will continue to rely on Combined Cycle Gas Turbines (CCGTs) and natural gas for her power demand. According to Second Minister for Trade and Industry S Iswaran, Singapore's net generation capacity will increase by more than 2000MW or about 20 percent of current installed capacity over the next few years and electricity will be generated by more efficient CCGTs (Wong, 2012). Power companies will increase the demand for more gas to fuel the new gas-fired generation capacity and thus meet a growing electricity demand in Singapore. BG Group has projected that Singapore's gas demand will nearly double to 16 billion cubic meters (565 billion cubic feet) in 2020 from 8.4 billion in 2010 (Koh, 2011). Singapore has increased its national gas consumption in electric power generation primarily for environmental reasons. Natural gas emits carbon dioxide and other pollutants when combusted but less than coal and oil.

Singapore imports piped natural gas from Indonesia and Malaysia, two of the largest gas exporters in the Asia Pacific region. Both countries are experiencing falling supply from ageing fields and rising domestic natural gas demand. They have been seeking a solution to the domestic gas shortage by starting to import liquefied natural gas (LNG) while continuing to export contracted LNG to countries like Japan and South Korea (International Energy Agency, 2011a). Malaysia's first liquefied natural gas LNG re-gasification facility, a 3.8 million tonnes per annum (mtpa) facility located near Sungai Udang Port in Malacca, was completed in June 2012 (LNG World News, 2012). The second terminal will be built in Johor by 2020 (Teo, 2012). For Indonesia, a three mtpa floating storage and re-gasification unit (FSRU) in the Jakarta Bay is its first LNG terminal out of the three which are being planned to be built (Reuters, 2012). Hence, Indonesia and Malaysia have both completed constructing their first LNG terminals this year and are expected to begin importing LNG later in the year.

In Singapore, the issue of high reliance on Indonesian and Malaysian gas has raised energy security concerns and inspired energy policies aiming at diversification of energy sources. The Republic's first LNG terminal is an effort to diversify supply of natural gas since it will enable the Republic to import

from a different number of countries. The terminal will thus reduce the country's dependence on Indonesian and Malaysia gas (Energy Market Authority, 2012). The facility will be completed in the second quarter of 2013 with a throughput or regasification capacity of 3.5 millions of tonnes per annum (mtpa). It will initially become operational with a regas plant, a jetty and two storage tanks. This capacity will increase to 6 mtpa by the end of 2013 when a third storage tank, two additional jetties and additional regas facilities are completed by the first quarter of 2014 (Energy Market Authority, 2010). There is also flexibility to expand its capacity by building up to four additional storage tanks and two LPG tanks.

UK's BG Group Inc. has a license to import LNG and sell regasified LNG in Singapore for up to 3 mtpa until 2023. BG has already sold 2.65 mtpa of LNG and is expected to reach 3 mtpa by 2013. Six power generation companies – Senoko Energy, PowerSeraya, Tuas Power Generation, SembCorp Cogen, Keppel Merlimau Cogen and GMR Energy – have committed to purchase 2 mtpa of regasified LNG (Energy Market Authority, 2010). Singapore is now exploring its options for importing more than 3 mtpa of LNG before the BG contract expires in 2023 (Gostelow, 2011).

With a terminal specifically designed to handle both imports and exports, LNG activities are expected to grow tremendously. In addition, Singapore is also strategically positioned between key LNG importers like China, India, Japan and South Korea and key LNG exporters like Qatar, Malaysia, Indonesia and Australia. In anticipation of growth in LNG-related activities, the Government of Singapore Investment Corp and China Investment Corp have pumped in US\$1 billion into Cheniere Energy's Sabine Pass terminal, the only terminal in the US to have received full approval to export LNG (Alfred, 2012). Singapore's Temasek Holdings have also invested US\$468 million for a 19.8% stake in Cheniere Energy and are in talk with Cheniere Energy to set up a marketing joint venture to import LNG to Asia from the Sabine Pass terminal (Lim, 2012).

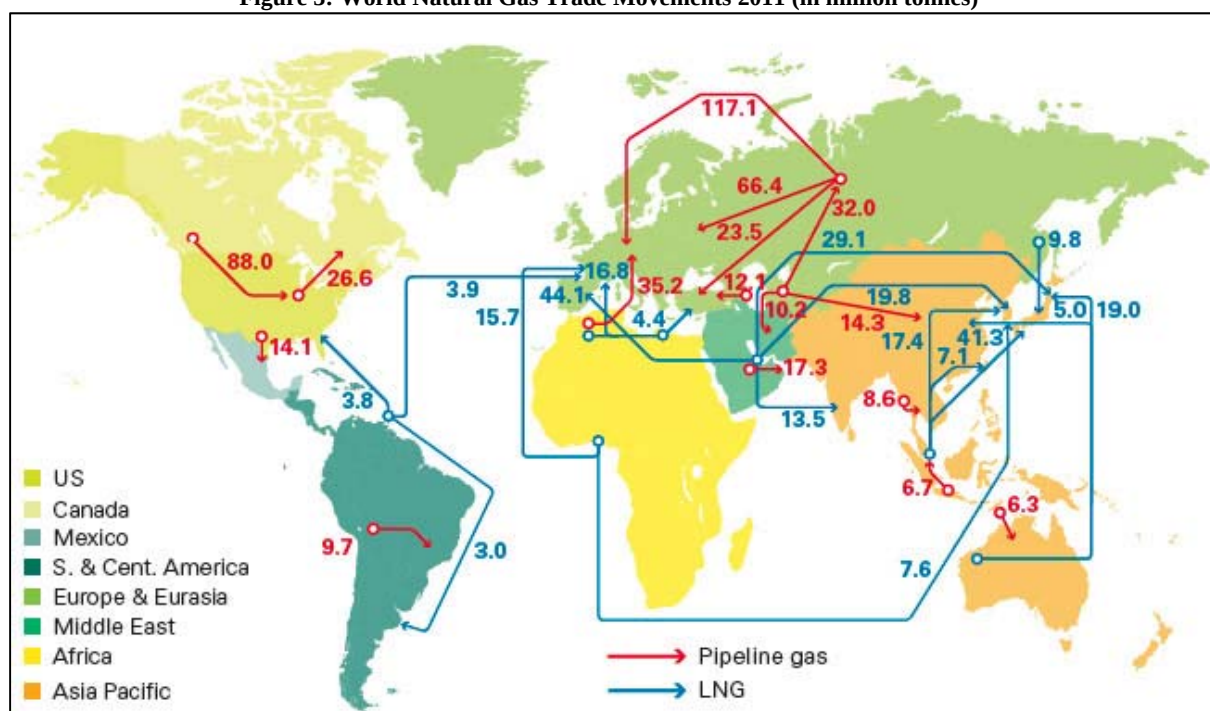
In summary, the SLNG terminal will allow Singapore to diversify imports beyond Indonesia and Malaysia. Increased availability of LNG from the massive expansion in LNG production in Qatar and Australia coupled with the rise in unconventional gas production in US will provide an abundant

source of alternative supply. This will help Singapore to meet her energy needs while allowing further diversification possibilities. In term of environmental impact, the carbon dioxide abatement potential for US LNG exports to Singapore will be limited given 80% natural gas in the power generation mix and limited natural gas vehicle expansions (Tan, 2011).

3.2. Price Reduction via Increase of Supply

The Asian liquefied natural gas (LNG) market is imperfectly competitive. Asia-Pacific countries such as China, Japan, India, Singapore, South Korea and Taiwan import LNG from a limited number of countries with market power which can charge high prices. As shown by Figure 3 below, supply is dominated by LNG imports from within the region such as Malaysia, Indonesia and Australia and from outside the region such as Qatar and Nigeria. Most LNG sold in the Asian market is under long-term contracts needed to finance the costs from constructing liquefaction facilities. The long term contract uses pricing formulas which links LNG prices to oil prices and the consumers are tied to these prices for the duration of the contract. Asian LNG buyers have been paying the highest worldwide prices for natural gas in recent years because the price of crude oil has also been high.

Figure 3: World Natural Gas Trade Movements 2011 (in million tonnes)



Source: BP Statistical Review of World Energy, 2012 from <http://www.bp.com/extendedsectiongenericarticle.do?categoryId=9041232&contentId=7075237>.

In contrast, US natural gas prices are market based and are determined by demand and supply interactions rather than through pricing formulas. Henry Hub (HH) is the benchmark of US natural gas price. If all the proposed US LNG export projects were operational, the United States would compete with Qatar in term of global export capacity. However, it must be noted that most of the proposed LNG export terminals are located on the US Gulf Coast and the East Coast. It means that US LNG will travel longer distance to reach Asia Pacific as compared to Qatar and Australia LNGs. When the project to widen the Panama Canal is completed by 2014, a new passage for LNG shipments from the US Gulf Coast to Asia will be opened up. This could make US LNG exports relatively more competitive with Middle Eastern and Australian LNG exports to the Asia Pacific.

Competition to supply the Asian market is likely to decrease Asian gas prices and thus weaken the Asian LNG price premium. The increased availability of LNG will have a positive impact on the liquidity of global LNG markets and help to serve the demand in Asia Pacific. Alaskan and West Canadian exports may be a source of strong competition for US LNG exports, given their shorter distances and thus lower transportation costs to the Pacific Basin. In West Canada, three export facilities namely Kitimat LNG, BC LNG and Texada Island LNG have applied for permits to export domestically produced natural gas as LNG with a cumulative capacity of almost 2.7 billion cubic feet per day (bcf/day) (Ratner et al., 2011). Australia is expected to have around 12 bcf/day export capacity by 2020 which is expected to almost exclusively serve Asia Pacific market (Ebinger et al., 2012). Thus, with projected growth in demand for natural gas in Asia expected to be competitively met by incremental supply around the world, Asian gas prices are likely to fall.

Existing producers have the option of reducing the prices they charge to Asian buyers to price US LNG exporters out of the market while continuing to maintain their share of the market. For example, Qatari gas producers reduced their price demand from Japan despite the post-Fukushima rise in Japan's LNG demand in response to the growing threat of Australia competition (Doshi et al., 2012).

The US gas supply push may not be able to change the use of oil-indexed pricing formulas overnight. The Asian market is still largely dependent on long term contracts. For example, recent contracts with

emerging new liquefaction capacity (like Australia) still adopt oil-indexed pricing (Ebinger et al., 2012). Strategic choice from the existing producer with a market power like Qatar will determine how pricing arrangements may evolve in the future (International Energy Agency, 2012). Qatar has the options of either undercutting the prices offered by other exporters, including from the US, on international markets and retaining or expanding export volumes, or alternatively maintaining higher prices for natural gas exports and losing market share. Nevertheless, a greater variety of natural gas sources brought about by the emergence of the US as an exporter for LNG could eventually challenge the very basis of Asian LNG pricing. Prices of US LNG based on hub indices establish alternate prices which are likely to be lower than prices based on traditional oil-indexed formulas.

Historically, Singapore like other Asian countries has paid the highest prices to guarantee supply. US LNG exports provide a potentially lower-price source of gas for Singapore, which has been highly dependent on Malaysia and Indonesia for its natural gas supplies. They could also give Singapore the option to either actively revise oil-indexed prices by renegotiating its natural gas contracts or increasingly explore these alternative pricing mechanisms. Following the Fukushima nuclear accident, for instance, Japan has been seeking new arrangements for LNG contracts to meet their increasing natural gas demand. Two of the contracts recently agreed, between Cheniere and GAIL (India) (Ramakrishnan, 2011), between Cheniere and KOGAS (Korea) (Reuters, 2012) have prices indexed to Henry Hub. The contract signed by BG Group, a British energy company which agreed to buy LNG from Cheniere to supply Asian and European markets, also adopts Henry Hub pricing (McAllister, 2011).

Therefore, US LNG exports will weaken the market power of incumbent LNG providers and put pressure on them to lower their natural gas prices, either by modifying oil-indexed pricing formulas to offer better terms or by delinking their export prices from oil. In particular, this could increase the negotiating power of the Singapore and reduce its natural gas import prices, as well as provide it with a source of LNG that is relatively cheap. Natural gas prices normally increase in case of an oil price shock because most Asian gas contracts are oil-indexed. The diversification possibility afforded by US LNG (which is not linked to oil markets) will also guard Singapore against the risk of oil price

increase. In general, additional volumes of US LNG will help to move the natural gas toward global market and hence price convergence.

4. Barriers to trans-Pacific LNG trade

Regulatory hurdles and Henry Hub price risk factors might impede transpacific gas trade. Growing US natural gas production has decreased the demand for both pipeline and LNG imports and shifted industry interest from building LNG import terminals to constructing LNG export terminals. Regulatory hurdles, infrastructure constraints and other factors could prevent LNG projects in the US from taking off and thus pose challenges in getting the natural gas to markets where it can be consumed, thus balancing global supply and demand.

4.1. Regulatory Approval

Projects to export liquefied natural gas (LNG) have been proposed and developed but none of the projects has reached a final investment decision (FID). Most projects are still evaluated on a case-by-case basis and waiting for regulatory approval. In the US, Lake Charles LNG, Cameron LNG, Freeport LNG, Cove Point LNG and Jordan Cove LNG terminals have received approval to construct liquefaction facilities and to export LNG to countries having Free Trade Agreement (FTA) with the US. Only the Sabine Pass LNG terminal has received full approval from DOE's Office of Fossil Energy to export LNG to both free-trade and non-free-trade countries. These approved liquefaction projects have an estimated total capacity of almost 9.54 billion cubic feet per day (bcf/day).

Table 1: Proposed North American LNG Export Projects

Terminal	Location	Volume (bcf/day)	FTA Approved	Non-FTA Approved
Sabine Pass	Louisiana	2.2	Yes	Yes
Freeport	Texas	1.4	Yes	Pending
Lake Charles	Louisiana	2.0	Yes	Pending
Carib Energy	Various	0.04	Yes	Pending
Dominion Cove	Maryland	1.0	Yes	Pending
Jordan Cove	Oregon	1.2	Yes	Pending
Cameron	Louisiana	1.7	Yes	Pending
Gulf Coast	Texas	2.8	Pending	Pending
Cambridge	Various	0.27	Pending	NA

Source: Energy Security Initiative at Brookings Report, 2012 from http://www.brookings.edu/~media/research/files/reports/2012/5/02%20lng%20exports%20ebinger/0502_lng_exports_ebinger.pdf.

Two regulatory approvals, namely an export authorization from the Department of Energy's (DOE) Office of Fossil Energy and a facility construction authorization from Federal Energy Regulatory Commission (FERC), are required. Issuance of an export authorization by DOE is based on a

judgment as to whether exports are consistent with the public interest. In addition, adding liquefaction capacities and modifying existing facilities require lengthy authorization process from FERC. FERC approvals are subject to compliance of regulatory, safety and environmental requirements. DOE and FERC will determine how fast and how far export approvals proceed depending upon the impacts on the public and the environment.

4.2. High US Natural Gas Prices

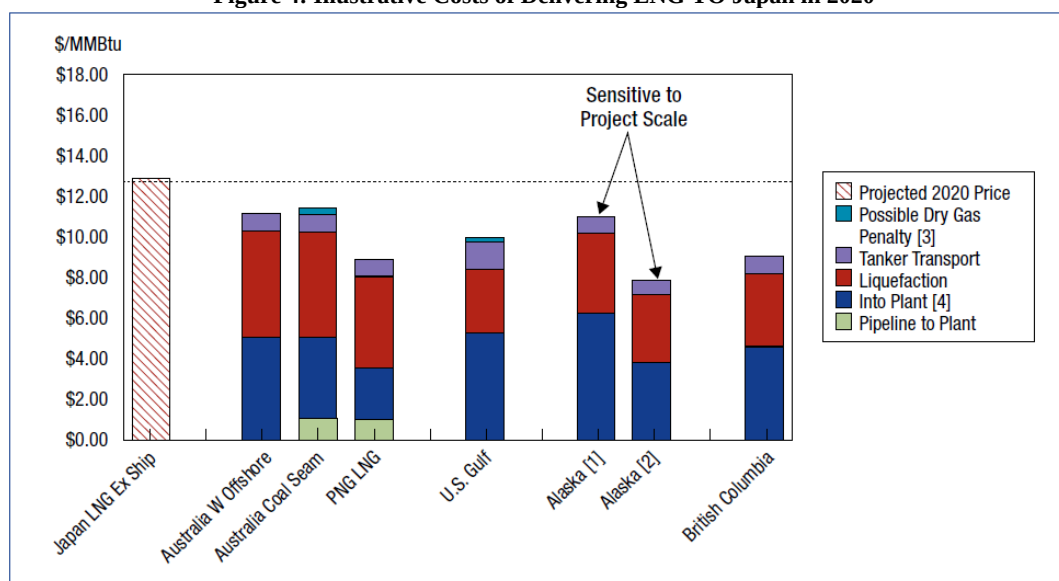
In the US, there is a growing debate as to the merits of exporting greater quantities of US natural gas. Firstly, domestic natural gas consumers fear that supplies will fall and prices will rise with greater US natural gas exports. Increase in domestic natural gas prices will lead to higher prices for businesses and households. Proponents and developers of natural gas export projects disagree, arguing that there will be ample supply to meet domestic demand and there will be added benefits from access to new overseas markets and higher international prices like increased revenues for government and greater domestic employment.

One key argument against exporting US natural gas is that low prices and large resources should increase the domestic use of the natural gas in the electricity generation, transportation and petrochemical sectors. Environmental concerns regarding emissions from coal power plants have spurred the growing share of natural gas in US power generation mix. Nevertheless, fuel switching requires huge costs and infrastructural developments so domestic increasing natural gas usage in other sectors outside electric power generation is not viable at this time. At this moment, it is unclear whether there will be a huge switch to natural gas consumption domestically so the magnitude of any rise on prices cannot be predicted with certainty.

Secondly, environmental groups have raised concerns over the use of hydraulic fracturing to produce shale gas because the production process contaminates water, produces emission and causes noise and seismic activity. Some groups are in favor of exporting natural gas since the potential rise in domestic gas prices will make renewable energies more viable while other groups argues that natural gas still emits greenhouse gases and switch to renewables should be done directly.

The proposed LNG export projects require private financings and interested buyers. Costs of producing, processing and shipping natural gas have placed a limit on the amount of LNG exports which are economic. The introduction of new regulations and legislations restricting shale gas development and LNG exports will negatively impact the economics of the proposed LNG export projects and provide a barrier to additional exports of LNG from the United States. The Energy Information Administration (EIA) and the International Energy Agency (IEA) project that the prices of natural gas in 2020 will be US\$5.18/MMBtu (Energy Information Administration, 2011) and US\$12.90/MMBtu (International Energy Agency, 2011b) for US and Japan markets respectively. Figure 3 illustrates that if Henry Hub prices increase so that the price differentials are less than around \$6/MMBtu, US LNG exports may no longer be economically feasible. However, it would require a substantial jump in Henry Hub prices for this to happen.

Figure 4: Illustrative Costs of Delivering LNG TO Japan in 2020



Source: James T. Jensen, 2012 from http://www.eia.gov/naturalgas/workshop/pdf/Session1_Jensen.pdf.

5. Conclusion

Gas demand in Asia Pacific is expected to grow in the next decade given the projected economic growth in China and India and assuming Japan's nuclear capacity remains offline. In the United States (US), exporting liquefied natural gas (LNG) in greater quantities is technically feasible given current breakthroughs in unconventional gas production, information on US natural gas resources and the forecasted supply growth. US LNG exports are competitive given the large price differential between US gas prices and oil prices, the basis for gas prices in Asia Pacific.

Increased US LNG exports have potential positive impacts on Singapore through supply diversity and price reduction by the possibility of reassessment of contractual pricing agreements which link natural gas and crude oil prices. Singapore can purchase LNG at lower prices by either entering into long-term supply contracts indexed to HH gas prices or by taking advantage of spot LNG prices when they are lower than oil-indexed prices.

Projects to develop liquefaction facilities will enable exports of LNG in increased quantities subject to approval by US policymakers. The extent of the approval is largely dependent upon the impact of exports on US domestic gas prices and domestic industrial sector. Introduction of new regulations and legislations restricting shale gas development and LNG exports could negatively impact the economics of the proposed LNG export projects and hinder additional exports of LNG from the United States.

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